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Shak M.S. Karim, Ph.D.

Atmospheric Scientist

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Professional Summary

Atmospheric scientist with 9+ years of experience in numerical modeling, atmospheric data analysis, and multiscale weather processes. Currently develops and leads the operational transition of a 100-member, 2-km convection-allowing ensemble for 48-h severe-weather prediction for Texas. Strong background in building and operating real-time and satellite-assimilated forecasting workflows, and processing large atmospheric datasets from NWP, reanalysis, radar & satellite remote sensing observations. Experienced in HPC environments, workflow automation, forecast product development, and engaging operational forecasters and emergency managers.

SKILLS

Tools and Languages	Python, MATLAB, Fortran, Bash scripting, Slurm, cron, Linux, Git, Conda, LaTex, QGIS, NCL, GrADS, NOAA's Weather and Climate Toolkit.
Numerical Modeling	WRF-ARW, NSSL WoFS-WRF, MPAS-A, HYSPLIT, WRF-LES, WRF-UCM, WRF-FIRE, CM1.
Forecasting Systems	Convection-allowing ensemble design and operations, real-time forecasting workflows, satellite-assimilated forecasting, fault-tolerant workflow automation, model configuration, probabilistic forecast product generation, web-based forecast visualization.
Data & Analysis	Model validation & verification, Large Dataset Analysis (GFS, ERA5, HRRR, RAP); weather radar calibration, measurements & analysis; spectral analysis; uncertainty quantification; statistical metric analysis.
Computing	Cloud Computing, HPC workflows (NCAR Cheyenne, Derecho, SWRC Pirate, TTU HPCC Nocona and ASU Tornado).

EDUCATION

- **Ph.D. in Atmospheric Science**, NC A&T State University, Greensboro, NC — Aug 2017 – Dec 2021
- **M.S. in Physics**, University of Chittagong, Bangladesh — Jul 2012 – Apr 2014
- **B.Sc. (Hons.) in Physics**, University of Chittagong, Bangladesh — Mar 2007 – Jun 2012

EXPERIENCE

NWP Research Scientist — NWI, Texas Tech University — July 2026 – Present
Ensemble Prediction System Development for Texas — Lubbock, TX

- Develop and maintain the Texas Ensemble Prediction System (TEPS), a 100-member, 2-km convection-allowing WRF ensemble producing 48-h severe-weather forecasts for Texas.
- Lead the move of TEPS from a fully validated retrospective system to real-time operations, designing the operational driver, run scheduling, and Slurm/cron job orchestration on HPC.
- Built TEPS products and deployed the product website, covering ensemble-mean, probabilistic and neighborhood-probability fields, paintball and plume displays, soundings, member-level views etc.
- Manage end-to-end production of the 100-member ensemble and its terabyte-scale output on HPC clusters (Tornado).
- Present TEPS products and capabilities to NWS forecasters, emergency managers, and partner institutions.

Postdoctoral Research Associate — SWRC, Hampton University — Jun 2024 – July 2026
Severe Weather Forecasting & Wind Retrieval Algorithm Development — Hampton, VA

- Operate and upgrade a satellite-assimilated severe weather forecasting system for real-time applications.
- Develop forecast-oriented workflows using Python, Bash, Slurm, and HPC resources.

- Perform high-res LES simulations to study boundary layer turbulence and convection.
- Investigating how the Urban environment controls post-landfall dissipation of Hurricanes.
- Develop a Doppler radar wind retrieval algorithm to examine turbulent wind structures in convection.
- Quantify uncertainty and analyze error metrics in retrieved wind fields.
- Operate the Skyler-II phased-array radar system for high-res observation of severe weather.

Postdoctoral Research Associate — NC A&T Jun 2022 – May 2024
Multi-scale Modeling of Urban and Tropical Weather Processes Greensboro, NC

- Developed multi-scale numerical modeling frameworks to study urban impacts on city convection.
- Analyzed how urban and Lake Michigan affect the precipitation patterns in Chicago.
- Identified key atmospheric dynamics not captured in NWS forecasts during the Yarnell Hill Fire entrapment of 19 firefighters.
- Investigated the forcing mechanisms of AEWs preceding the genesis of the Atlantic hurricane.
- Examined the Moist Flow Regime Transition and Heavy Orographic Rainfall during Typhoon's passing over Taiwan's CMR.

Adjunct Faculty — NC A&T Jan 2022 – May 2024
Teaching & Advising Student Research Projects Greensboro, NC

- Taught undergraduate and graduate courses, including Atmospheric Modeling and Atmospheric Physics.
- Developed an atmospheric modeling course for senior-year students, introducing them to state-of-the-art numerical modeling techniques.
- Mentored Atmospheric Science Undergraduate Students' research projects, enhancing students' academic and professional development.

Research Assistant — NC A&T Aug 2017 – Dec 2021
Fire Weather Modeling & TC Track Dynamics Greensboro, NC

- Conducted numerical simulations to study multiscale atmospheric processes.
- Investigated orographic effects on producing severe downslope windstorms during wildfires.
- Applied HYSPLIT trajectory modeling to analyze atmospheric transport processes.
- Developed a conceptual model describing stages of Santa Ana wind-driven wildfire evolution.
- Investigated the dynamics & physical process of tropical cyclone track deflection.
- Analyzed the moist convection process leading to heavy rainfall passing over the 3D isolated mountains.

Research Intern — USDA Forest Service Jun 2019 – Aug 2019
Wildfire Modeling with Future Climate Scenario Greensboro, NC

- Designed WRF simulations with RCP climate scenarios to assess future wildfire risk.
- Investigated climate change impacts on wildfire-conducive mesoscale environments.

HONORS and AWARDS

- 2021 NCAT Outstanding Senior Graduate Teaching Assistant Award (2020–2021 academic year)
- Outstanding Poster Presentation Award, MSCAR (September 2020).
- Grant to attend Unidata workshop by University Corporation for Atmospheric Research (March 2020)
- 2019 Doctoral Summer Intern Fellowship – USDA Forest Service & NCAT (April 2019)
- Sweden-Bangladesh Trust Fund Fellowship for pursuing PhD in the USA (2018)
- University of Chittagong Merit Scholarship for being the 1st in Bachelor of Science (2012)
- Exim Bank Foundation Scholarship for undergraduate study (2008 - 2012)

Grant and Proposal Contributions

- **NSF AI Datasets Flagship Proposal [In Preparation]**
  Title: *An integrated Texas weather, social and economic dataset for AI (working title)*
  Principal Investigator: Brian Hirth
  Organization: Texas Tech University
  Contribution: Co-writing the ensemble component: TEPS ensemble curation and retention, and observation–model pairing.

- **NSF Proposal** – 2609729 **[Not Funded]**
  Title: *Collaborative Research: Understanding and Quantifying the Dissipation of Tropical Cyclones through Realistic Urban Environments*
  Principal Investigator: Shak Md Shajedul Karim
  Organization: Hampton University
  Amount: $411,601
  Submitted: December 5, 2025
  Contribution: Led and contributed to the development of a collaborative NSF proposal focused on understanding how realistic urban environments influence tropical cyclone dissipation, including research planning, scientific framing, methodology development, and proposal writing.

- **NSF Postdoctoral Research Award** – AGS-1900621
  Title: *Examining the HDW Fire Weather Index by Conducting WRF-SFIRE Simulations of Wildfires that Occurred over the Mountains in the Southwest U.S.*
  Supervisor: Dr. Yuh-Lang Lin
  Amount: $83,664
  Period: January 1, 2022 – December 31, 2022

PUBLICATIONS [Complete List]

8. Sub-Grid Discoveries from Phased Array Radar Observations (2026): Mackey T., Guimond S., **Karim S.M.S.** *Bulletin of the American Meteorological Society, Under Review.*
7. A Ground-based Multi-Doppler Wind Retrieval Algorithm for Turbulent Convection: An LES-Based Radar Simulator Framework (2026): **Karim et al.** *Remote Sensing*, 18(15), 2468. doi:10.3390/rs18152468.
6. Urban Impacts on Convective Squall Lines over Chicago in the Warm Season – Part II: A Numerical Study of Urban Infrastructure Effects on the Evolution of City-Scale Convection (2025): **Karim et al.** *Atmosphere*, 16(6), 652. doi:10.3390/atmos16060652
5. Urban Impacts on Convective Squall Lines over Chicago in the Warm Season – Part I: Observations of Multi-Scale Convective Evolution (2025): Kaplan M.L., **Karim S.M.S.**, Lin Y.L. *Atmosphere*, 16(3), 306. doi:10.3390/atmos16030306
4. Convective Density Current Circulations that Modulated Meso-γ Surface Winds Near the Yarnell Hill Fire (2023): Kaplan M.L., **Karim S.M.S.**, Wiles J., James C.N., Lin Y.L., Riley J. *Fire*, 6, 130. doi:10.3390/fire6040130
3. Formation Mechanisms of the Mesoscale Environment Conducive to a Downslope Windstorm over the Cuyamaca Mountains Associated with Santa Ana Wind during the Cedar Fire (2022): **Karim et al.** *Journal of Applied Meteorology and Climatology*, 61(11), 1797–1818. doi:10.1175/JAMC-D-22-0025.1
2. The Multi-Scale Dynamics Organizing a Favorable Environment for Convective Density Currents Preceding the Yarnell Hill Fire (2021): Kaplan M.L., James C.N., Ising J., Sinclair M.R., Lin Y.L., Taylor A., Riley J., **Karim S.M.S.**, Wiles J. *Climate*, 9(12), 170. doi:10.3390/cli9120170
1. A Study of Track Deflection Associated with the Landfall of Tropical Cyclone Sidr (2007) over the Bay of Bengal and Bangladesh (2021): **Karim et al.** *Dynamics of Atmospheres and Oceans*, 93, 101207. doi:10.1016/j.dynatmoce.2021.101207

CONFERENCE PROCEEDINGS & TALKS [Selected]

17. Ground-based dual-Doppler Radar Algorithm Development and Satellite Data Assimilation for Severe Weather Applications (Sep 2025): **Karim S.M.S.** *Invited TALKS at NASA LARC.*
16. Parallel Approaches Towards Radar Algorithm Development and Satellite Data Assimilation for Severe Weather Applications (Mar 2025): **Karim S.M.S.** *SEMINAR at Center for Atmospheric Science, HU.*
15. Developing a Ground-based Doppler Radar Algorithm to Compute 3D Winds – A LES Model Simulator Approach (Aug 2025): **Karim et al.** *41st International Conference on Radar Meteorology.*
14. Study the Moist Flow Regime Transition and Heavy Orographic Rainfall During Typhoon Passage over

Taiwan's Central Mountain Range (May 2024): **Karim et al.** *36th AMS Hurricane.*

13. Investigate the Forcing Mechanisms Leading to the Formation of the African Easterly Waves Preceding the Genesis of Hurricanes in the Atlantic Ocean (Jul 2023): **Karim et al.** *20th AMS Mesoscale.*
12. Formation of Tropical Storm Debby (2006) due to thermal and mechanical forcing associated with the passage of pre-Debby African Easterly Waves passing over the Guinea Highlands (July 2023): Wiles JT, **Karim S.M.S.** *20th AMS Mesoscale.*
11. Study of the Dynamics and Upstream Track Deflection Associated with Typhoon Nepartak (2016) Passing over Taiwan's CMR (Jul 2023): Graham R, **Karim S.M.S.**, Lin YL, *20th AMS Mesoscale.*
10. Moist Flow Regime Transition and Heavy Orographic Rainfall Formation Mechanisms During Super Typhoon Netpartak's Passage over Taiwan's Central Mountain Range (May 2023): **Karim et al.** *ICMCS-XV.*
9. Convective Density Current Circulations that Modulated Meso-γ Surface Winds Near the Yarnell Hill Fire (May 2023): Kaplan M.L., **Karim S.M.S.**, Wiles J.T., James C.N., Lin Y.L. *14th AMS Fire.*
8. Upstream Mountain Ranges' impacts on the Formation of the Cedar Fire on the Lee Slope of the Cuyamaca Mountains (Jul 2022): **Karim et al.** *20th AMS Mountain.*
7. Complex Interactions Between Convective Density Currents, Terrain Heating, Channeling, and Blocking Leading to Erratic Wildfire motions (Jul 2022): Kaplan M.L., **Karim S.M.S.**,... *20th AMS Mountain.*
6. Orographic Impacts on Upstream Track Deflection and Heavy Rainfall Associated with Typhoon Nepartak (2016) over Taiwan's Central Mountain Range (July 2022): **Karim et al.** *20th AMS Mountain.*
5. The Multi-Scale Dynamics Organizing a Favorable Environment for Convective Density Currents that Redirected the Yarnell Hill Fire (Jan 2022): Kaplan M.L., **Karim S.M.S.**,... *102nd AMS Annual Meeting*
4. Formation Mechanisms of the Mesoscale Environment Conducive to the Cedar Fire (May 2021): **Karim et al.** *13th AMS Fire and Forest Meteorology Symposium*
3. Mesoscale Dynamics Resulting in the Formation of the Cedar Fire (2003) on the lee slope of the Mountains (Sep 2020): **Karim et al.** *MSCAR.* **(Outstanding Poster Award)**
2. A Study of Track Deflection Associated with the Landfall of Tropical Cyclone Sidr (2007) on Bangladesh (Oct 2019): **Karim et al.** *10th Annual Hurricane Conference.*
1. Impacts of Hurricane Maria (2017) on Puerto Rico (Apr 2019): Samara S., **Karim S.M.S.**, Lin YL, *NC A&T Undergraduate Research Symposium.* **(Second-Best Poster Award)**

WORKSHOPS

- NWI Operational Flooding Tools Workshop (presenter, TEPS forecast products) — Oct 2026, National Wind Institute, Texas Tech University, Lubbock, TX
- Joint Field Experiments on Heavy Rainfall Workshop — May 2023, CSU, Boulder, CO
- Prediction of Rainfall Extremes Campaign in the Pacific (PRECIP) Science workshop — May 2023, CSU
- NASA's Science Mission Directorate (SMD) Wildfire Stakeholder Engagement Workshop — Feb 2022
- Satellite Data Application Training Short Course — Jul 2021, NOAA (online)
- joint WRF and/or MPAS Users online workshop — Jun 2021, NCAR MMM Lab
- Virtual MPAS-A tutorial — Apr 2021, NCAR Mesa Lab
- Fire Weather Research Workshop — Apr 2021, SJSU (online)
- Unidata Users Workshop on Python — Mar 2021, UCAR
- Technical Workshop on Severe Weather Forecasting — Sep 2020, UIUC
- HYSPLIT workshop — Jun 2020, NOAA (online)
- Cloud computing for WRF and initialized WRF using HRRR data — NCAT 2019

Professional Service & Mentoring

- **Peer Reviewer:**
  - Quarterly Journal of Royal Meteorological Society (QJRMS)
  - Springer Nature
  - Proposal Reviewer - 7th International Fire Behavior and Fuels Conference

  - NWA's Annual Meeting.
- **Judge:** 36th AMS Hurricane Conference; NC A&T Undergraduate Research Symposium.
- **Supervised Students:**
  - Sumer Samara (Undergraduate Student, 2018–2019, NC A&T)
  - Nicklaus Duncan (Undergraduate Student, 2022–2023, NC A&T)
  - Richarde Graham (Undergraduate Student, 2023–2024, NC A&T)
- **Event & Volunteering:** Energy Day Event (2018); Virginia Tornado Summit Event (2025); Conduct WRF Tutorial (Fall 19, 21, 22; Spring 21) at NC A&T.

CERTIFICATIONS & TRAINING

- Tropical Cyclone Rainfall - Forecasting and Communication
- Forecasting Tropical Cyclone Storm Surge
- PBL in complex terrain
- Introduction to Ensembles: Forecasting Hurricane Sandy,
- Advanced Fire Weather Forecasters Course (COMET/UCAR)
- Assessing Fire Danger and Fire Weather Forecasting
- Fire Weather Climatology
- Mesoscale Meteorological Effects on Fire Behavior
- Topographic Influences on Wildland Fire Behavior
- Detecting and Monitoring Wildfires using GOES-R/JPSS
- Grid Techniques for Wildfire Modeling
- AMS Satellite Application Training